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A Mechanical Analysis of the Grab Starting Technique in Swimming

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The purpose of this study was to identify the mechanical characteristics of the hands-between-the-feet grab starting technique which contribute to a faster start. Twenty-four high school swimmers performed four trials of a grab start followed by a glide to a distance of 9 m. The results suggested that to obtain a faster start, swimmers should (a) move the center of mass fast in the forward direction while the feet are in contact with the starting block, (b) maximize the force exerted through the feet in the backward direction, and (c) maximize the force exerted through the hands against the starting block in the forward and upward direction.

In competitive swimming the goal is to cover a set distance in the least amount of time. This means that a swimmer must start, swim, and turn fast. Although the time spent starting is very small, the differences between winning and losing a race are often so small that this time can be decisive.

A few investigations have been conducted on this subject. In most cases (Bloom, Hosler, & Disch, 1978; Bowers & Cavanagh, 1975; Hanauer, 1967, 1972; Michaels, 1973) the results suggest that the grab start is more effective than other starting techniques. In other cases, however, (Ayalon, Van Gheluwe, & Kanitz, 1975; Lowell, 1975) the results suggest the opposite. Finally, there are still other cases (Gibson & Holt, 1979; Lewis, 1980) in which the results suggest that there are no differences between the grab start and other starting techniques.

Parts of this paper have been published previously in a modified form for the use of teachers, coaches and swimmers.

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Although there has been some dispute over which starting technique in swimming is most effective, the grab start has been the most widely used method of starting at all levels of competition for some years. Hanauer (1972) reported that all six finalists in the 50-yd freestyle at the 1971 National Collegiate Athletic Association (NCAA) Swimming Championships—that is, the United States championships for eligible university swimmers—used the grab start. Havriluk and Ward (1979) made a similar observation with respect to all competitors in the 50-yd freestyle at the 1976 NCAA Swimming and Diving Championships.

Observations suggest that swimmers perform the grab start with different mechanical characteristics. Thus, it seems rather important that the characteristics which lead to fast starts be identified.

The purpose of this study was to identify the mechanical characteristics of the hands-between-the-feet grab starting technique which contribute to a shorter starting time.

Procedure

Variables Measured

The objective in performing a swimming start is to get the fastest starting time possible. For the purposes of this study, the starting time was divided into: (a) the time from the starting signal until takeoff from the block—the *block time,* (b) the time the swimmer spent in the air—the *flight time,* and (c) the time from the first contact with the water until first contact with a bulkhead placed 9 m from the start—the *water time.* Each of these partial times is determined by a number of variables.

The block time (Figure 1) is determined by the horizontal and vertical displacements of the center of mass, CM (Figure 2) and by the average horizontal and vertical velocities with which these displacements occur. The horizontal and vertical displacements of the CM are determined by the initial and final x- and y-coordinates of the CM. (The initial x- and y-coordinates of the CM indicate the location of the CM at the instant the starting signal is given; the final x- and y-coordinates of the CM indicate the location of the CM at the instant of takeoff.)

The flight time (Figure 3) is determined by the three characteristics which determine the flight time of any projectile: (a) the vertical velocity at takeoff, (b) the height of the

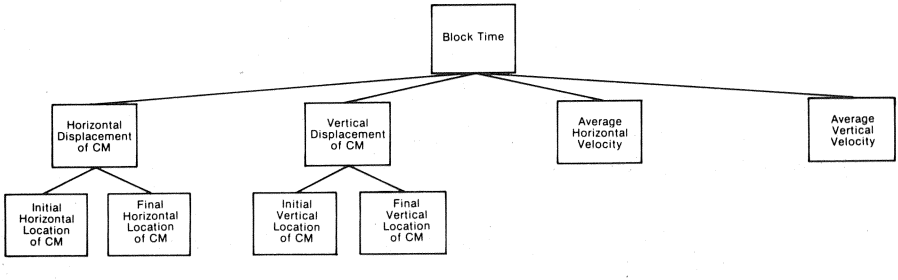

Figure 1 — Factors influencing the block time—that is, the time from the starting signal until takeoff from the block.

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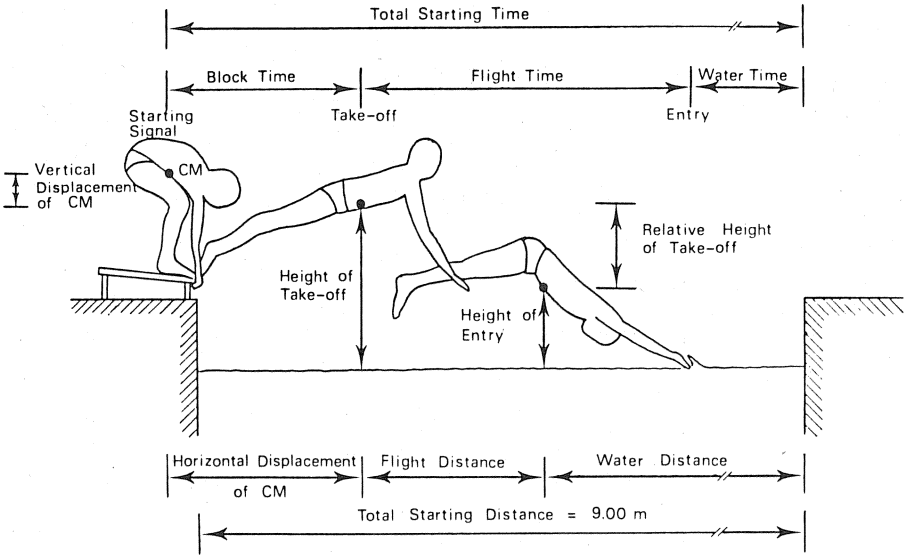

Figure 2 — Diagrammatic representation of selected variables.

CM at takeoff relative to its height at landing or entry, and (c) the air resistance encountered in flight. Since the vertical velocity is zero at the instant the starting signal is given, the vertical velocity at takeoff is determined by the resultant vertical impulse and the mass of the swimmer (impulse-momentum relationship). The resultant vertical impulse is determined in turn by the vertical impulses exerted on the swimmer's hands and feet. The vertical impulse on the hands is determined by the average vertical force exerted on the hands and by the time the hands are in contact with the front edge of the starting block. The

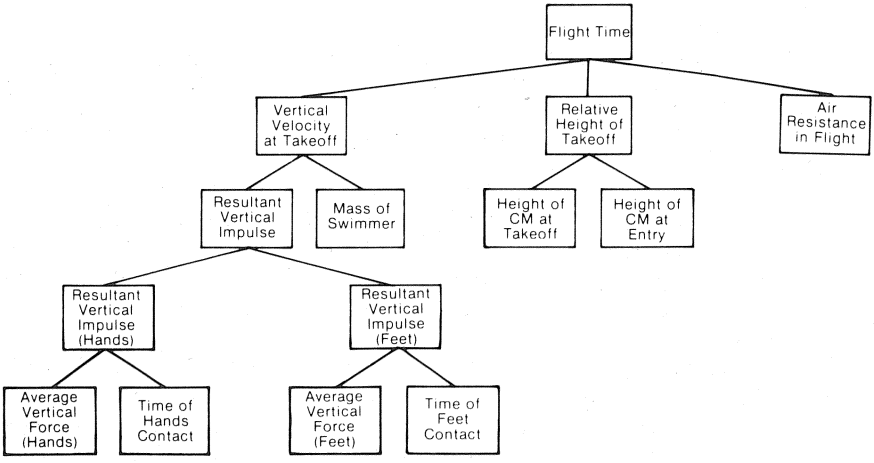

Figure 3 — Factors influencing the flight time.

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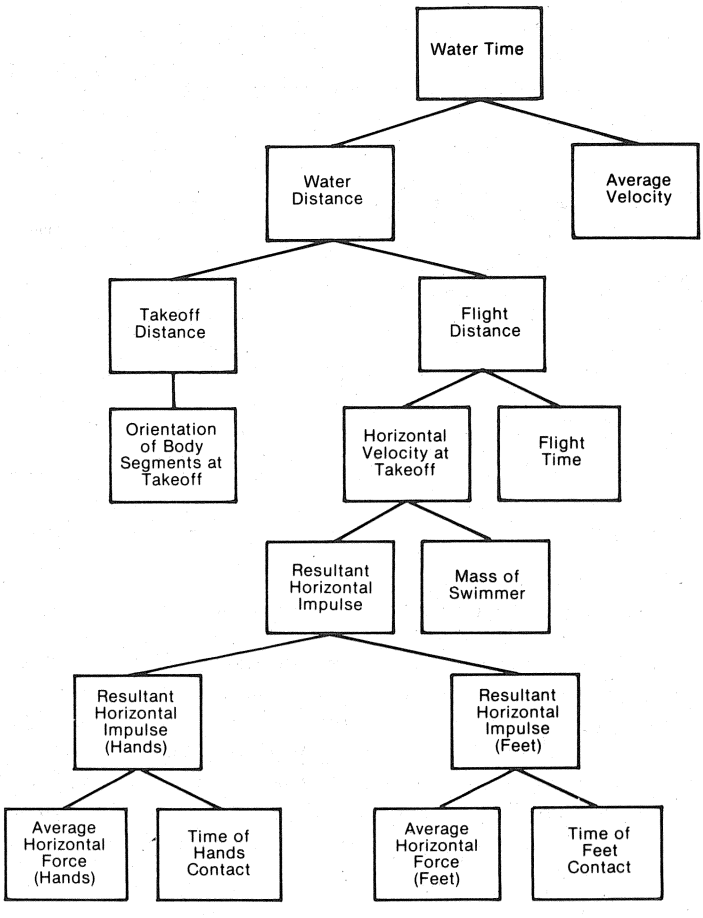

Figure 4 — Factors influencing the water time—that is, the time from first contact with the water until first contact with the bulkhead 9 m from the start.

vertical impulse exerted on the feet is similarly determined by the average vertical force exerted on the feet and by the time the feet are in contact with the starting block.

The relative height of the takeoff is determined by the height of the CM at takeoff and by the height of the CM at entry. The height of the CM at takeoff is determined by the inclination of the segments of the swimmer's body at takeoff and the height of the CM at entry by the inclinations of the segments at entry. The air resistance is considered to be negligible because of the characteristics of the motion (a relatively slow velocity of takeoff and a reasonably streamlined form).

The water time (Figure 4) is determined by the water distance—that is, the horizontal distance between the CM of the swimmer at entry and the bulkhead (Figure 2)—and by the average velocity of the swimmer over this distance. The water distance depends on the takeoff distance—that is, the distance of the CM from the start end of the pool at takeoff—and on the flight distance. The first of these depends on the orientation of the

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swimmer's body at takeoff. The second depends on the horizontal velocity at takeoff and the flight time. The horizontal velocity at takeoff is determined by the resultant horizontal impulse and the mass of the swimmer (impulse-momentum relationship). The resultant horizontal impulse, in turn, is determined by the horizontal impulses exerted on the swimmer's hands and feet. The horizontal impulse exerted on the hands is determined by the average horizontal force exerted on the hands and by the time the hands are in contact with the front edge of the starting block. The horizontal impulse exerted on the feet is similarly determined by the average horizontal force exerted on the feet and by the time the feet are in contact with the starting block. The factors that determine the flight time have already been discussed.

All of the variables referred to in the preceding paragraphs and shown in the deterministic models of Figures 1, 3, and 4 were measured in this study.

Subjects

Twenty-four male high school swimmers who were participants in an intensive swimming camp at the University of Iowa were used as subjects in this study. They had been involved in competitive swimming for an average of 6 years (range = 2 to 10 years). The mean of their best times for the 50-yd (45.72 m) freestyle was 24.05 s (range = 22.44 to 29.97 s).

Instrumentation

A measurement system consisting of (a) a starting and timing system, (b) cinematographic equipment, and (c) a force measuring device, was used to record the performance of each trial.

A Colorado Time System (Colorado Time Systems, 2050 14th St. S.E., Loveland, CO) was used to start and time the subjects up to the 9-m distance. This distance was selected on the basis of a pilot study designed to determine the maximum distance to which any of the subjects would glide under competitive conditions. The system consisted of a central unit which was electrically connected to a horn and to a touch-pad attached to the pool bulkhead. The horn was used to emit the starting signal. The timing system was turned on when the starting signal was given and turned off when the subject's fingers touched the touch-pad. The system provided a print out of the starting time accurate to 1/1000th of a second.

A 16-mm electrically driven Locam camera (Red Lake Labs, Inc., 2991 Corvin Dr., Santa Clara, CA) mounted on a stable tripod with the frontal plane of the lens set parallel to the plane of motion and 16.00 m distant from it was used to film the performance of each trial. The filming speed was set at 100 frames/s. A timing light generator pulsed a light-emitting diode mounted inside the camera to mark the edge of the film at constant intervals of 0.1 s.

The forces exerted by the subject's hands on the front edge of the starting block were measured with the aid of a simply-supported steel bar (diameter = 0.025 m) mounted on the front edge of the starting block. Strain gauges were bonded near both ends of the steel bar to indicate the deformation of the bar in the horizontal and vertical directions due to the forces exerted by the subject's hands. A rigid aluminum sheet was used to cover the top of the starting platform and to prevent contact between the subject's feet and the

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force bar. This sheet had a 0.25-m long exposed region between the subject's feet so that the hands could grip the force bar.

The force bar was connected in series to a bridge amplifier and to a MFE (MFE Corp., Keewaydin Dr., Salem, NH) three-channel strip-chart recorder. Two channels were used, one for the horizontal and one for the vertical direction. The attenuators for the horizontal and vertical direction channels were set at 10mV/mm which yielded a 0.01-m deflection for each 0.1 V signal emitted by the amplifier. The construction and original calibration of the force bar was discussed by Vaughan (1980).

Synchronization of Measurement System

A switch box was used to control the starting signal, a light bulb set in the plane of view of the lens and an event marker on the strip chart recorder. When the on/off switch was turned on, the starting signal sounded, the light bulb was switched off, and the event marker made a mark on the thermal paper of the strip-chart recorder.

Testing Procedures

The 24 subjects were divided into four groups of five and one group of four. All subjects within each group performed their trials during an interval of 1 hour. Each subject performed four trials. Each trial consisted of a grab start and glide until the subject's finger made contact with the touch-pad mounted on the pool bulkhead. The subjects were instructed to warm up and to perform starts as they would do in a 50-m freestyle event. Kicking and/or stroking was not allowed.

Film Analysis

The analysis of the films was conducted on a Vanguard Motion Analyzer (Vanguard Instrument Corp., 1860 Walt Whitman Rd., Melville, NY). Three randomly selected trials per subject were analyzed. (Note: A preliminary analysis of the data revealed that the inclusion of the fourth trial contributed nothing additional to the analysis.)

Impulses and Forces

The impulse-momentum relationship for the horizontal direction can be written in the form:

$$\int_{t_1}^{t_2} (F_x^f + F_x^h)\, dt = m\,(V_{G_x})_2 \qquad (1)$$

where t_1 and t_2 are the instants at which the starting signal was given and of takeoff, respectively; F_x^f is the forward force exerted on the subject's feet; F_x^h is the forward force exerted on the subject's hands; $(V_{G_x})_2$ is the forward component of the CM velocity at takeoff; and m is the subject's mass.

The corresponding equation for the vertical direction is:

$$\int_{t_1}^{t_2} (F_y^f + F_y^h - F_y^w)\, dt = m\,(V_{G_y})_2 \qquad (2)$$

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where F_y^f is the upward force exerted on the subject's feet; F_y^h is the upward force exerted on the subject's hands; F_y^w is the subject's weight; and $(V_{G_y})_2$ is the upward component of the center of mass velocity at takeoff.

The horizontal and vertical impulses exerted on the subject's hands were obtained by integrating the respective force-time curves obtained experimentally through the strain-gauged bar. A Numonics digitizer (Numonics Corp., 418 Pierce St., Lansdale, PA) was used to measure the area under each curve. The mass and weight of the subject were known. The horizontal and vertical components of the CM velocity at takeoff were obtained from film. Equations (1) and (2) could thus be solved, respectively, for the horizontal and vertical impulses exerted on the subject's feet.

Results

The means and standard deviations for the masses and heights of the subjects and the Pearson product-moment correlations of these quantities with the starting time are presented in Table 1. (The level of significance adopted in this study was $P \leqslant .05$.)

Table 1

Means and Standard Deviations for Subject Mass and Height, and Correlation for Subject Mass and Height With Starting Time

	M	*SD*	Correlation with starting time r
Mass (Kg)	64.47	7.89	−.69*
Height (m)	1.73	.09	−.74*

*$P \leqslant .05$

Among the variables measured, several could be influenced by the mass and/or height of the subjects. To overcome the problems inherent in this situation, each of the independent variables was correlated with the dependent variable (the starting time) with subject mass and height partialled out.

The means and standard deviations of all independent variables that were significantly correlated with the starting time are presented in Table 2.

Further partial correlations were computed to investigate the possible existence of significant interrelationships between selected independent variables. These partial correlations are presented in Figure 5. (Note: With the exception of the two cases referred to in Table 1, all correlations reported in this paper are partial correlations with subject mass and height partialled out.)

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Table 2

Means, Standard Deviations, and Partial Correlation Coefficients for Independent Variables Significantly Correlated (P≤ .05) With Starting Time

	M	*SD*	Correlation with starting time Partial r
Average horizontal velocity of subject's center of mass during block time (m/s)	1.42	.18	– .25
Resultant horizontal impulse (N.s)	254.94	45.06	– .24
Inclination of feet to forward horizontal at takeoff (deg)	75.80	9.20	.26
Height of entry (m)	.59	.08	– .32
Water time (s)	3.55	.87	.97
Average water velocity (m/s)	2.00	.38	– .84

Discussion

The only one of the three partial times significantly correlated with the starting time—the water time—indicated that shorter water times were associated with shorter starting times. In fact, the correlation between those two variables indicated that the water time accounted for 94% of the variance in the starting time.

The water time is completely determined by the water distance and the average water velocity. The water distance was found to vary very little from subject to subject and from trial to trial (SD = 0.22m) and was not significantly correlated with the starting time (r = .10). The average water velocity, on the other hand, was significantly correlated with the starting time (Table 2), and was clearly the dominant factor at this level of the analysis.

The average water speed depends logically on the horizontal velocity at entry and on the drag forces acting on the subject during the glide—the greater the horizontal velocity at entry and the less the subsequent drag forces, the greater the average water velocity. However, these two factors are basically incompatible. The drag force acting on a body moving through a fluid is proportional to the square of its speed:

$$F_D = C_D \varrho A V^2 / 2 \quad (3)$$

where F_D is the drag force; C_D is the coefficient of drag; ϱ is the fluid density; A is the frontal area of the body; and V is the velocity of the body. Thus, a large horizontal

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velocity at entry necessarily implies large opposing forces during the glide. A large horizontal velocity at entry is, in short, both a help and a hindrance. A nonsignificant correlation was found between the horizontal velocity at entry and the starting time ($r = -.18$). This suggests that a large horizontal velocity at entry is slightly more help than hindrance. (Note: For the purposes of computing the horizontal velocity at entry, it was assumed that air resistance was negligible. The horizontal velocity at entry was thus equated with the horizontal velocity at takeoff.)

The factors influencing the drag forces were not measured in this study. The focus was instead on those factors associated with the swimmer's movements on the block and in the air. The next step in the analysis consisted therefore of a consideration of those factors that determine the horizontal velocity at takeoff. This velocity is determined, in accord with the impulse-momentum relationship, by the mass and the resultant horizontal impulse. The finding of a significant correlation between the resultant horizontal impulse and the starting time (Table 2)—a correlation of similar magnitude to that between the horizontal velocity of takeoff and the starting time ($r = -.18$)—was thus exactly as might be expected.

The next step in the analysis consisted of examining the factors that determine the magnitude of the resultant horizontal impulse (I_x^r). This impulse is equal to the sum of the horizontal impulses exerted upon the hands (I_x^h) and the feet (I_x^f):

$$I_x^r = I_x^h + I_x^f \tag{4}$$

When I_x^h and I_x^f were each correlated with I_x^r, a significant correlation was found for I_x^f ($r = -.85$) but not for I_x^h ($r = 0.18$)—Figure 5. Although the first of these two results was expected, the lack of a significant relationship in the case of the I_x^h demanded a close examination of the role played by the hands in the grab start. The results showed that the hands generally pulled forward and upward against the block rather than pushing backward and downward as one might have expected. The horizontal forces that the block exerted on the hands thus acted to oppose the forward motion of the body as a whole.

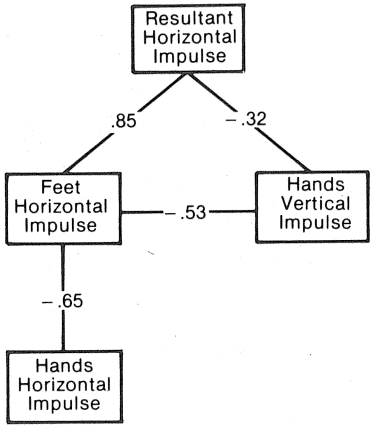

Figure 5 — Summary of results obtained for kinetic variables. Significant correlations ($P \leqslant .05$) are shown between linked variables.

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This finding and the suggestion that the function of the arm pull was to pre-tense and thus facilitate the drive from the legs—a suggestion made by Cavanagh, Palmgren, and Kerr (1975), who reported a similar finding for a single subject—led to the next step of the investigation. In this step, the relationships among the vertical impulse exerted upon the hands (I_y^h), the vertical impulse exerted upon the feet (I_y^f), and the I_x^r were determined. A significant negative relationship was found between I_y^h and I_x^r ($r = -.32$) indicating that the larger the downward I_y^h, the larger the I_x^r. No significant relationship was found between the I_y^f and the I_x^r ($r = .15$). These findings are summarized in Figure 5. Further significant correlations indicated that the larger the backward I_x^h and the larger the downward I_y^h, the larger the forward I_x^f. These findings thus supported the notion that the function of the arm pull is to facilitate the leg drive.

In addition to the variables already discussed, three others were found to be significantly correlated with the starting time (Table 2). The significant correlations involved in these cases indicated that faster horizontal speeds of the CM during block time, smaller foot inclinations at takeoff, and higher heights of entry were associated with shorter starting times.

The first of these three results might well have been expected. The second might be explained in terms of the final drive of the subject's feet against the starting block—a forceful plantar flexion of the ankles during this final drive producing both a fast start and a small inclination of the feet at takeoff. A significant correlation ($r = -.29$) between the foot inclination at takeoff and the feet horizontal impulse tended to support this suggestion.

The last of these three findings was not easy to explain. One's first inclination might be to suggest that the body position adopted in achieving a high position of the CM at entry was one that minimized the drag during the glide that followed and thus led to a reduction in the starting time. However, this suggestion was not supported by the finding of a nonsignificant correlation between the height of entry and the average water velocity ($r = .04$). Thus it appears that more work is needed to determine whether a high height at entry is a desirable characteristic of the grab starting technique.

Conclusions

On the basis of these findings it seems appropriate to suggest to swimmers who perform the grab starting technique with the hands between the feet that they should: (a) move their centers of mass fast in the forward direction while their feet are in contact with the starting block, (b) maximize the force exerted through the feet against the block in the backward direction, and (c) maximize the force exerted through the hands against the front edge of the starting block in the forward and upward direction.

References

Ayalon, A., Van Gheluwe, B., & Kanitz, M. (1975). A comparison of four styles of racing start in swimming. In J.P. Clarys & L. Lewillie (Eds.), *Swimming II*, pp. 233-240. University Park Press, Baltimore.

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Bloom, J.A., Hosler, W.W., & Disch, J.G. (1978). Differences in flight, reaction and movement time for the grab and conventional starts. *Swimming Technique* 15:34-36.

Bowers, J.A., & Cavanagh, P.R. (1975). A biomechanical comparison of the grab and conventional sprint starts in competitive swimming. In J.P. Clarys & L. Lewillie (Eds.), *Swimming II,* pp. 225-232. University Park Press, Baltimore.

Cavanagh, P.R., Palmgren, J.V., & Kerr, B.A. (1975). A device to measure forces at the hands during the grab start. In J.P. Clarys & L. Lewillie (Eds.), *Swimming II,* pp. 43-45. University Park Press, Baltimore.

Gibson, G., & Holt, L.E. (1979). A cinema-computer analysis of selected starting techniques. *Swimming Technique* 13:75-76.

Hanauer, E.S. (1967). The grab start. *Swimming World* 8:5, 42.

Hanauer, E.S. (1972). Grab start faster than conventional start. *Swimming World* 13:8-9, 54-55.

Havriluk, R., & Ward, T. (1979). A cinematographic analysis of three grab starts. *Swimming Technique* 16:50-52.

Lewis, S. (1980). Comparison of five swimming starting techniques. *Swimming Technique* 16:124-128.

Lowell, J.C. (1975). Analysis of the grab start and the conventional start. *Swimming Technique* 12:66-69, 79.

Michaels, R.A. (1973). A time distance comparison of the conventional and the grab start. *Swimming Technique* 10:16-17.

Vaughan, C. (1980). An optimization approach to closed loop problems in biomechanics. Doctoral dissertation, University of Iowa, pp. 141-150.

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